MILITARY SPECIFICATION SHEET

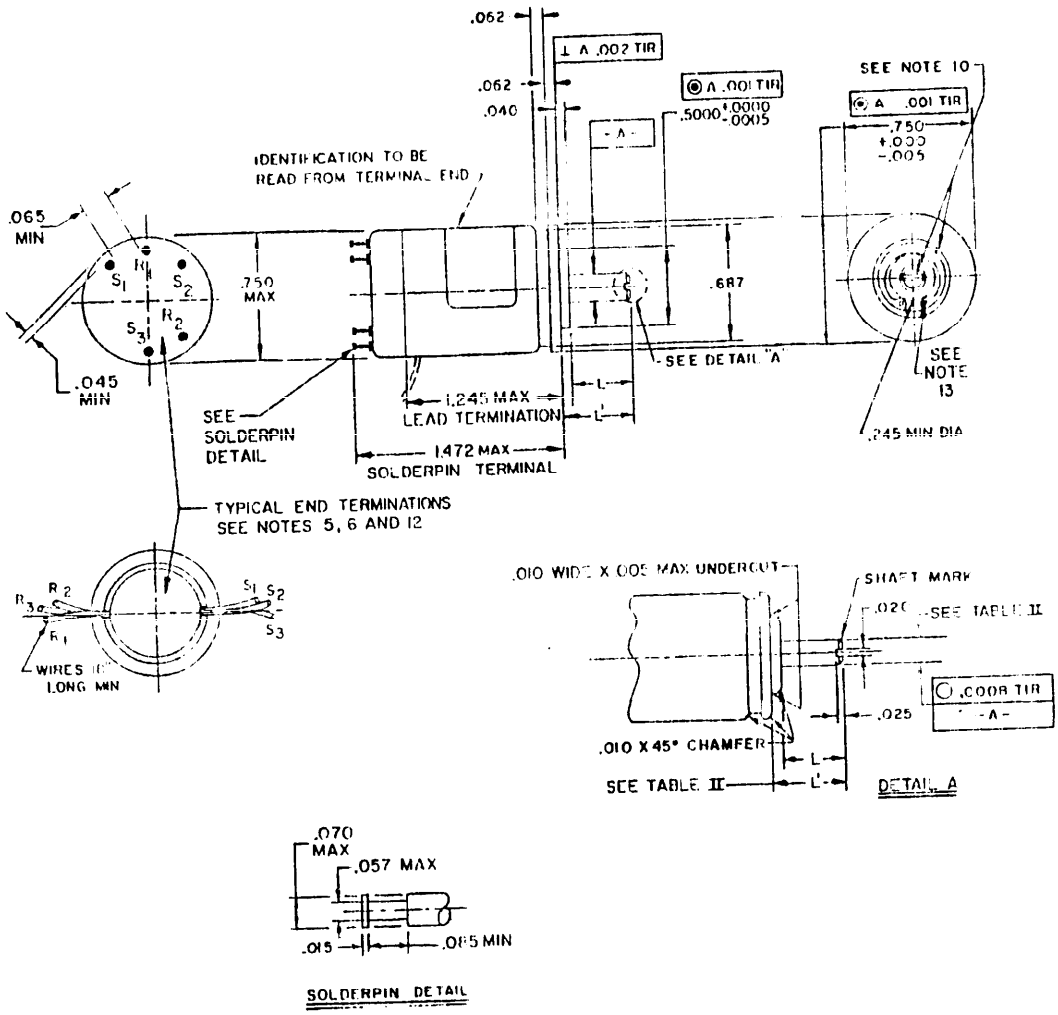
SYNCHRO, TORQUE DIFFERENTIAL

TRANSMITTER 26V-08TDX4a

(THESE SYNCHROS ARE FOR REPLACEMENT PURPOSES ONLY; NOT RECOMMENDED FOR NEW DESIGNS.)

The complete requirements for procuring the synchro described herein shall consist of this document and the latest issue of MIL-S-20708.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



FSC 5990

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NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals, ±2° on angles.
3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, then the housing rotated.
4. Leads shall be type E-28.
5. Terminal lead wires shall depart through the back surface of the terminal end of the synchro in any acceptable manner. The terminal pins shall be marked on the unit as indicated.
6. Minor variations of unit configuration are permitted for undimensioned design detail.
7. Wire lead stress shall be performed as a qualification test.
8. Solder pin terminal stress shall be tested in accordance with MIL-STD-202, Method 211 as a qualification test. The terminals shall be capable of withstanding, without damage or loosening, a steady pull of the force specified in any direction by a wire fastened in the normal manner. Internal connections to terminals shall be welded or soldered with high temp solder, SN-5 per QQ-S-571.
9. Pending issuance of a qualified products list (QPL) a sample of the synchros to be furnished under this specification shall be subjected to the preproduction tests as specified in MIL-S-20708.
10. Mark on housing is to match shaft mark within 10° when rotor is set at approximately synchro zero and will bear no relation to the centerlines shown on this drawing.
11. Housing material shall be of corrosive resistive material, e.g., stainless steel or aluminum.
12. Decals are permitted for terminal designation.
13. The front retainer shall be threaded or a snap ring.

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TABLE I

SYNCHRO TORQUE DIFFERENTIAL

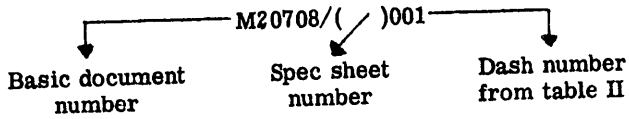
TRANSMITTER 26V-08TDX4a

PERFORMANCE REQUIREMENTS AND CHARACTERISTICS

Requirement	Value	Unit	Tolerance
	Stator	---	---
1. Excitation	400	Hz	±1%
2. Frequency	11.8	Volts	±1%
3. Primary voltage	91	Milliamps	Maximum
4. Primary current	.21	Watts	Nominal
5. Primary power			
6. D.C. Resistance:			
Rotor	36 (Reference)	Ohms	---
Stator	24 (Reference)	Ohms	---
7. Impedance:			
Zso	28 +j116	Ohms	R±15%, +jx±10%
Zrs	47 +j13 (Reference)	Ohms	---
Zro	35 +j125	Ohms	R±15%, +jx±10%
8. Transformation ratio	1.154	---	±2%
Output/input	9 (Lead)	Degrees	±1.5
9. Phase shift	7.0	Minutes	Maximum
10. Electrical error			
11. Null voltage:			
Total	30	Mv rms	Maximum
Fundamental	20	Mv rms	Maximum
12. Friction torque:			
@ 25° C	.04	Oz-In	Maximum
	(3)	(Gm-Cm)	
@ -55° C and +125° C	.12	Oz-In	Maximum
	(9)	(Gm-Cm)	
13. Radial play	.0004 (Using 4 oz load)	Inches	Maximum
14. End play	.0003-.0008 (Using 8 oz load)	Inches	Min-Max
15. Terminal stress	2	Pound	$^{+2}_{-0}$ Oz
(See note 8)			
16. Temperature rise	20	°C	Maximum

Part number: The part number consists of the basic number of this specification sheet and a dash number taken from table II.

Example:



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TABLE II

Dash No.	Shaft diameter (+.0000) (-.0005)	Shaft style	Termination	Shaft length	
				L (ref)	Dim. to first L' mounting flange surface (±.010)
-001	.0900	Plain	Wire leads	.085	.125
-002	.0900	Plain	Wire leads	.250	.290
-003	.0900	Plain	Wire leads	.335	.375
-004	.0900	Plain	Wire leads	.375	.415
-005	.0900	Plain	Wire leads	.435	.475
-006	.0900	Plain	Wire leads	.500	.540
-007	.0900	Plain	Wire leads	.585	.625
-008	.0900	Plain	Wire leads	.625	.665
-009	.0900	Plain	Solder pins	.085	.125
-010	.0900	Plain	Solder pins	.250	.290
-011	.0900	Plain	Solder pins	.335	.375
-012	.0900	Plain	Solder pins	.375	.415
-013	.0900	Plain	Solder pins	.435	.475
-014	.0900	Plain	Solder pins	.500	.540
-015	.0900	Plain	Solder pins	.585	.625
-016	.0900	Plain	Solder pins	.625	.665
-017	.1200	Plain	Wire leads	.085	.125
-018	.1200	Plain	Wire leads	.250	.290
-019	.1200	Plain	Wire leads	.335	.375
-020	.1200	Plain	Wire leads	.375	.415
-021	.1200	Plain	Wire leads	.435	.475
-022	.1200	Plain	Wire leads	.500	.540
-023	.1200	Plain	Wire leads	.585	.625
-024	.1200	Plain	Wire leads	.625	.665
-025	.1200	Plain	Solder pins	.085	.125
-026	.1200	Plain	Solder pins	.250	.290
-027	.1200	Plain	Solder pins	.335	.375
-028	.1200	Plain	Solder pins	.375	.415
-029	.1200	Plain	Solder pins	.435	.475
-030	.1200	Plain	Solder pins	.500	.540
-031	.1200	Plain	Solder pins	.585	.625
-032	.1200	Plain	Solder pins	.625	.665

Custodians:
Army - MU
Navy - AS
Air Force - 85

Review activities:
Army - EL, MI, MU, AT, AV, ME
Navy - SH, AS
Air Force - 11, 17, 85

User activities:
Army -
Navy - SH, MC, CG
Air Force -

Preparing activity:
Air Force - 85

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